Attributing Non-Direct Speech, Thought, and Writing to Characters

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Introduction

Speech, thought, and writing representation (STWR) encompasses various techniques through which fictional characters' voices, thoughts, and written communications are integrated into a narrative text. These modes of STWR play a central part of many narrative theories (e.g., Leech and Short 2007, Martínez and Scheffel 2019), and are typically categorized into four main forms: direct STWR (exact quotations; "Then he thought: *I'm hungry*"), indirect STWR (paraphrased content; "She asked *where the food was*"), free indirect STWR (character's voice mediated through the narrator; "He was at a loss. *Where should he ever find lunch here?*"), and reported STWR (mentioned communications; "She wrote *about the lunch*").

This talk presents first results of ongoing work on automatic detection of STWR in German narrative text and attributing them to speakers, extending upon previous work (e.g., Su et al. 2024, Vishnubhotla et al. 2023) that focuses on *direct* speech only. The primary goal is to investigate how different presentation techniques of STW are employed by authors for specific fictional characters along certain dimensions, e.g., characters' and authors' gender. Similar to research on gender visibility in fiction (Kraicer and Piper 2019), emotions in gendered characters (Schumacher and Flüh 2020), character agency and gender (Jockers and Kirilloff 2016, Viane et al. 2023), or character descriptions (Underwood et al. 2018) we want to examine which characters "get to speak" and how authors present this speech.

For training and evaluation, the REDEWIEDERGABE corpus (Brunner et al. 2020) was employed, which annotated STWR in German news texts and fiction from c. 1840–1920 (488K tokens). Each STWR span is annotated with a type (direct, indirect, free indirect, reported), and a speaker span. The speaker span is on the textual surface level, usually covering a pronoun of the speaking character.

We divide the task of STWR recognition into two token classification sub-tasks: First, the *detection* of STWR instances, identifying spans with their respective type, and second, the *attribution* of STWR, where, given a specific STWR instance in context, annotate the speaker mention tokens.

While many corpora with direct speech annotation exist (e.g. DROC for German, Krug et al. 2017; PNDC for English, Vishnubhotla et al. 2022), REDEWIEDERGABE is one of few literary corpora including free indirect and reported STWR, next to the English STOP corpus (Semino and Short 2004). However, STOP lacks speaker mention spans. This is why this study primarily focuses on German language.

Method & Results

Given above training data, two different Transformer-based systems were trained: (a) A state-of-the-art encoder-decoder model (mT5-large, Xue et al. 2021)[1] to jointly perform the two tasks of detection and attribution, fine-tuned to "translate" the input to pseudo-XML output.[2] Such structured generation has proven successful in a similar-fashioned STWR task on German news texts (Petersen-Frey and Biemann 2024). (b) Two BERT-based encoders (XLM-roBERTa-large, Conneau et al. 2020) fine-tuned for detection resp. attribution using sequence tagging. Both systems are evaluated on a test set of narrative texts from the REDEWIEDERGABE corpus (42k Tokens; overall Fleiss' kappa of the annotations is 0.73).

Surprisingly, GPT-4o proved challenging to prompt effectively. Even with extensive prompt engineering and few-shot examples, GPT-4o could not outperform above BERT system (Speech/Speaker CEAF F1 of 0.35/0.14).

Figure 1 shows the performance of the different systems on the test set, grouped by *type*, reporting raw classification performance next to CEAF metrics (Luo 2005) on STWR instances, where the latter metric aligns system outputs with gold references.

Model	mT5-large			XLM-roBERTa-large full inference			XLM-roBERTa-large w/ gold speech	
Speech Type	Speech Token F1	Speech CEAF F1	Speaker CEAF F1	Speech Token F1	Speech CEAF F1	Speaker CEAF F1	Speaker CEAF F1	Num Instances
direct	0.79	0.83	0.57	0.93	0.76	0.63	0.78	446
indirect	0.73	0.68	0.49	0.79	0.64	0.62	0.91	125
reported	0.56	0.51	0.38	0.67	0.57	0.48	0.79	219
free indirect	0.34	0.44	0.14	0.78	0.48	0.21	0.35	40
all	–	0.71	0.49	–	0.68	0.57	0.78	830

Figure 1: Performance on the REDEWIEDERGABE test set.

Both models demonstrate satisfactory results in STWR recognition. Free indirect STWR is recognized worst, likely due to limited training examples. The systems also struggle with reported STWR, which was challenging even for human annotators (Fleiss' kappa 0.49), having often blurry boundaries with pure narration and indirect STWR (Brunner et al. 2020). BERT outperforms the other model in most aspects, except for direct STWR instances, where it struggles with extended passages. These results, combined with the low performance of GPT-4o, add to accumulating evidence that small fine-tuned models can often compete with more resource-intensive ones (e.g., Bamman et al. 2024). As such, BERT should not be disregarded and still remains an affordable, fast, effective, and accessible method for DH, especially when high-quality training data is available.

Additionally, the models also showed potential for zero-shot cross-lingual transfer learning: while trained on German data only, they successfully annotated examples in English, French, and Italian. However, a thorough evaluation would require high-quality evaluation data in these languages.

Proof of Concept

As a proof of concept, I annotated the German novel *Effi Briest* using above BERT models, combining the results with existing gold coreference annotations to attribute STWR to specific characters. Figure 2 shows the distribution of direct, indirect, and reported speech for each character with ≥20 STWR instances (excluding free indirect STWR due to low prevalence). For now, this analysis limits its notion of characters' gender to that marked by their pronouns ("he/him" and "she/her")[3].

While STWR presentation techniques vary considerably between characters, particularly along the *direct vs. reported* axis, only subtle gender-based patterns can be observed, with "he/him" characters showing a slight tendency toward more reported and less direct STWR. However, these differences are possibly confounded by the characters' narrative functions.

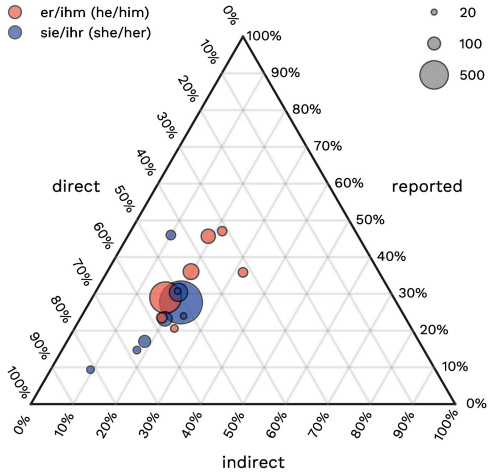

Figure 2: Distribution of STWR type per character in *Effi Briest*.

Outlook

Starting with the exploratory proof of concept, for future work, we want to extend the investigation with broader theoretical foundations, a larger corpus, and rigorous methodological validation. Next to this, we plan to continuously improve above models by incorporating more training data (e.g., direct speech corpora), and to build a multilingual test set to quantitatively assess applicability to other languages.

Notes

1. mT5-xl was also fine-tuned (using LoRA), but did not outperform mT5-large. This is in line with negative results by Petersen-Frey and Biemann (2024).
2. E.g., <speaker>She<1></speaker> asked where the food was<indirect/><1>, for indirect speech, and <1> connects speaker and speech.
3. Of course, this binary conceptualization is insufficient to capture the full extend of gender, and, e.g., ignores performative dimensions; here however, we heuristically follow the approach of *referential gender* like in BookNLP, as we are primarily concerned with conventional roles of characters (cf. Underwood et al., 2018).

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